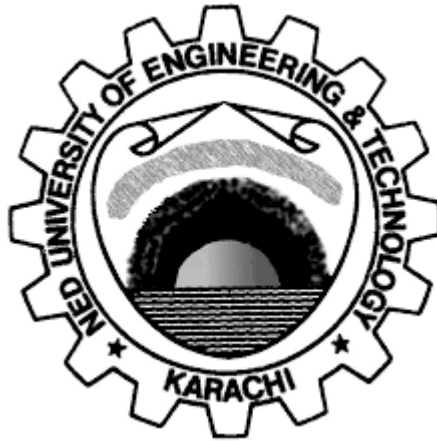


Complex Engineering Problem Report

Electrical power System Protection EE-457

Final Year-Electrical Engineering

Batch: 2019



Group Members:

Syed Hassan Ali Rizvi (EE-19135)

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Submitted to: Dr. Abdullah Muneer

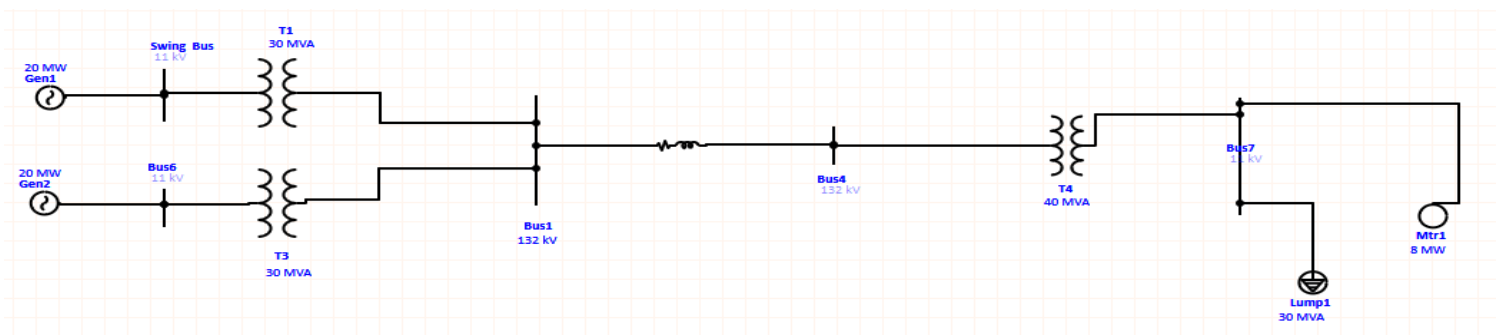
Submission Date: 23/1/2023

System details:-

The system we proposed consists of 5 buses and contains two generators, three transformers, 1 lump load and 1 induction motor load.

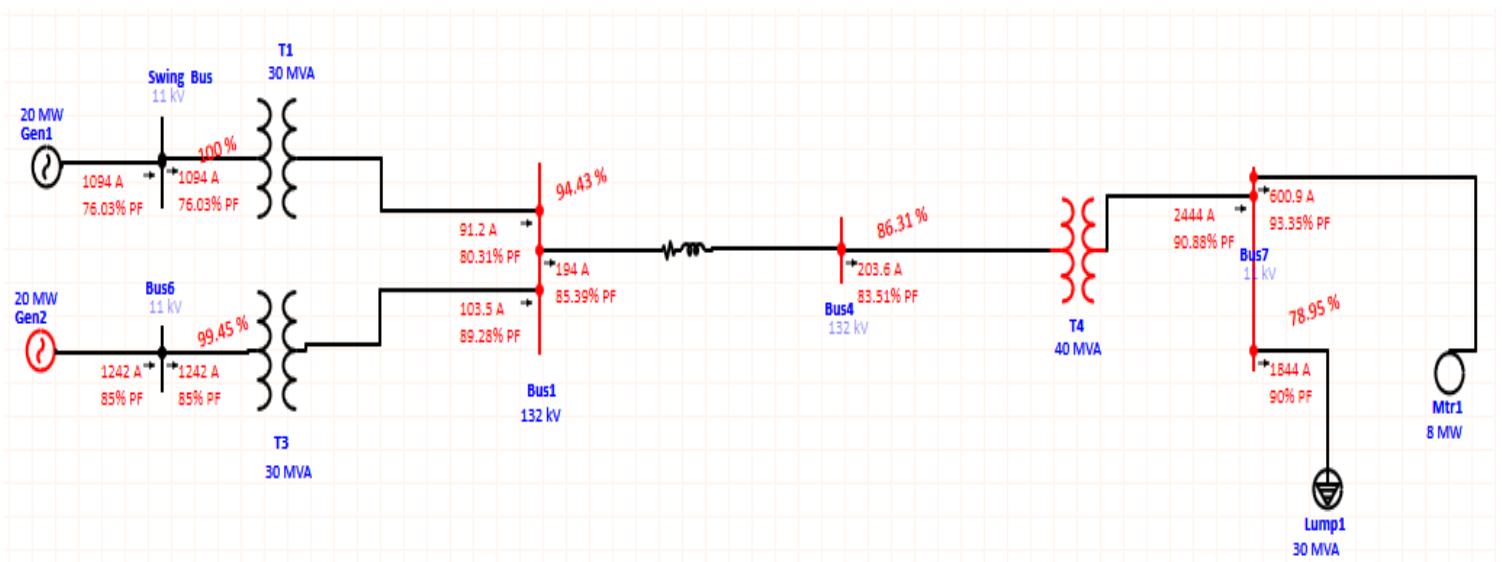
Instrument	Rating
1). Generator Set 1 (Gen 1)	20 W
2). Generator Set 2 (Gen 2)	20 W
3). Transformer T1	11/132 kV 30 MVA
4). Transformer T2	11/132 kV 30 MVA
5). Transformer T3	132/11 kV 40 MVA
6). Load	Lumped 30 MVA & I/M Motor 8 MW
7). Transmission line	100 km long KRYPTON and GND wire is TIGER

Schematic:-



Load Flow Analysis:

The load Flow was perform for maximum of 100 iterations and Newton Raphson method is used.



Following, the report regarding load flow is attached:-

Project: EPSP CEP

ETAP

Page: 1

19.0.1C

Date: 21-01-2023

Contract: None

SN:

Engineer: Syed Hassan Ali Rizvi

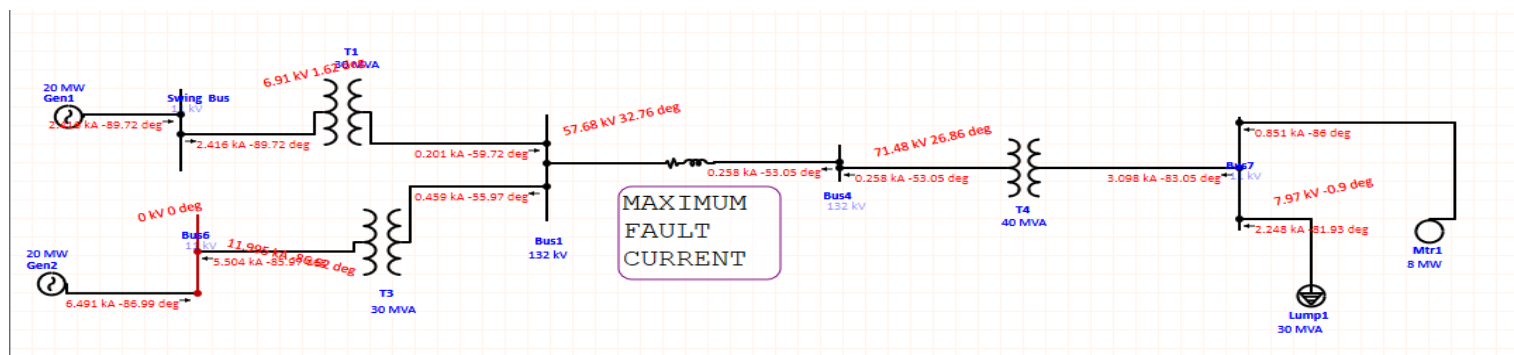
Revision: Base

Filename: EPSPCEP

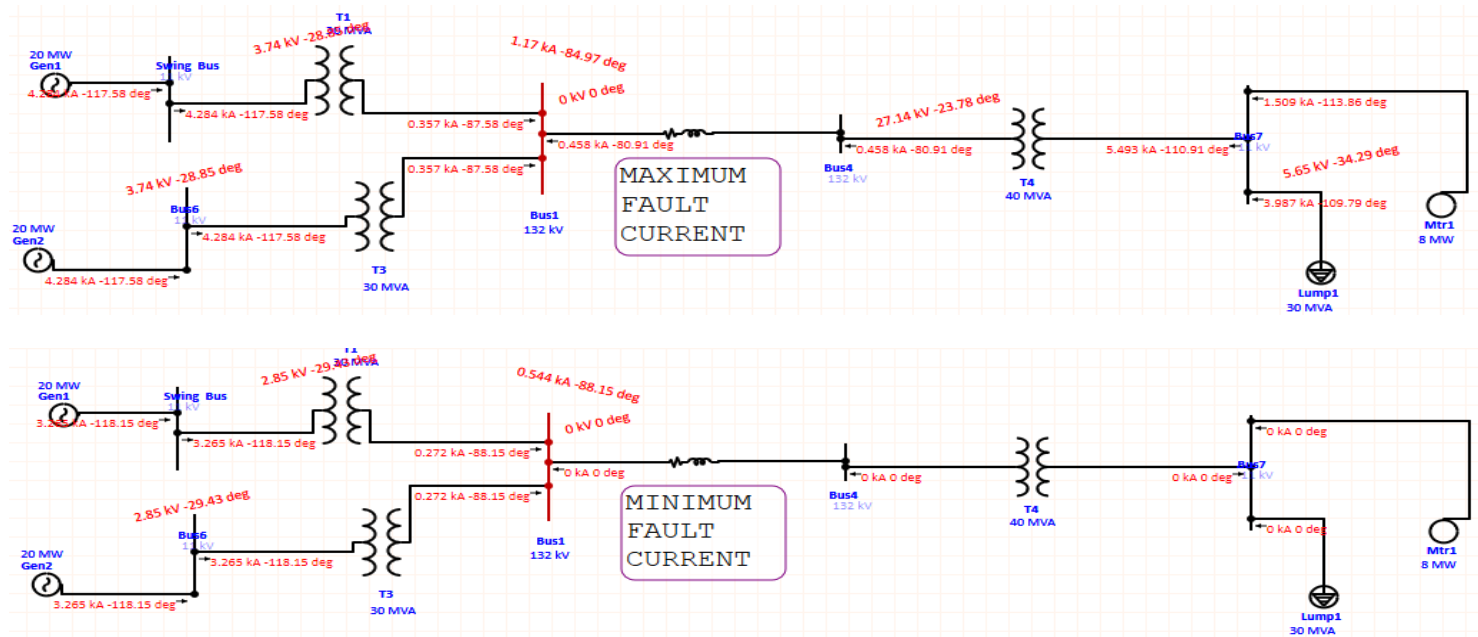
Config.: Normal

LOAD FLOW REPORT

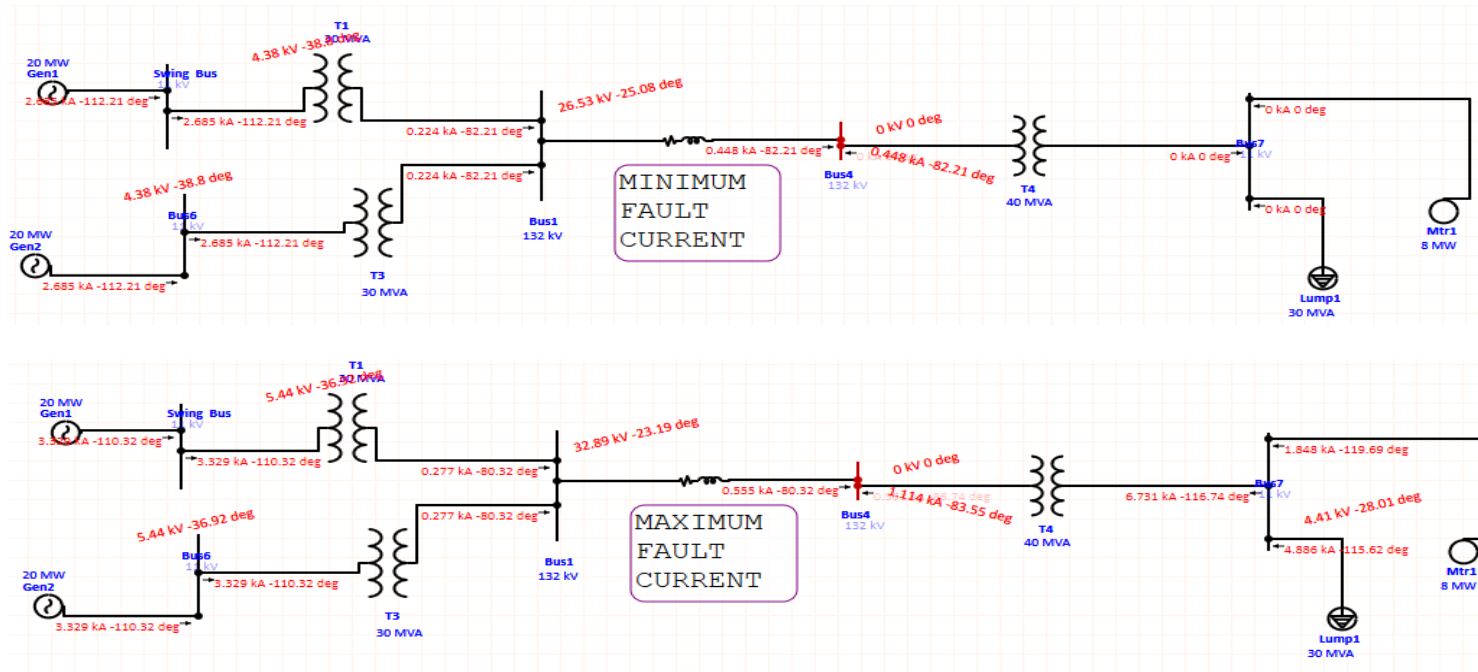
Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Bus1	132.000	94.434	-3.9	0.000	0.000	0.000	0.000	Bus4	35.762	21.796	194.0	85.4	
								Swing Bus	-15.813	-11.733	91.2	80.3	
								Bus6	-19.948	-10.063	103.5	89.3	
Bus4	132.000	86.308	-6.3	0.000	0.000	0.000	0.000	Bus1	-33.554	-22.104	203.6	83.5	
								Bus7	33.554	22.104	203.6	83.5	
Bus6	11.000	99.450	1.1	20.000	12.395	0.000	0.000	Bus1	20.000	12.395	1241.8	85.0	
Bus7	11.000	78.949	-15.0	0.000	0.000	33.404	15.333	Bus4	-33.404	-15.333	2443.5	90.9	
Swing Bus	11.000	100.000	0.0	15.854	13.544	0.000	0.000	Bus1	15.854	13.544	1094.4	76.0	



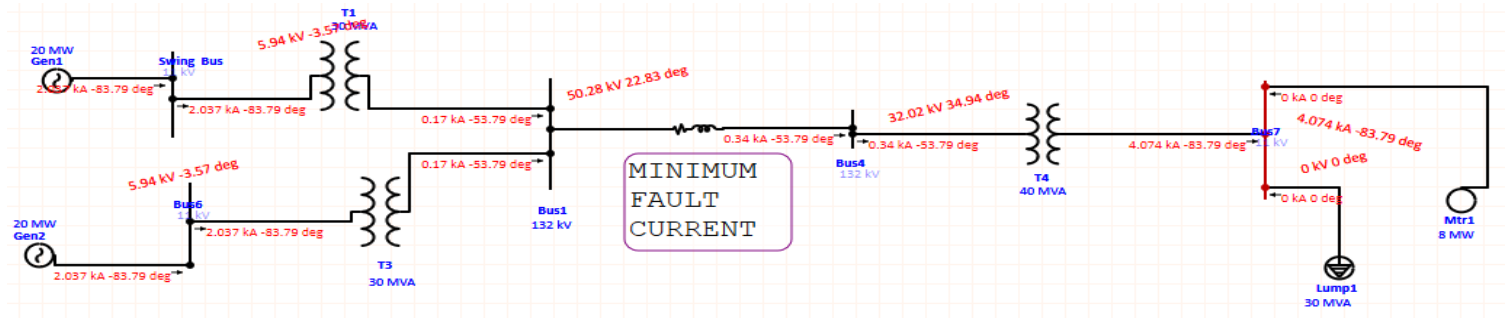
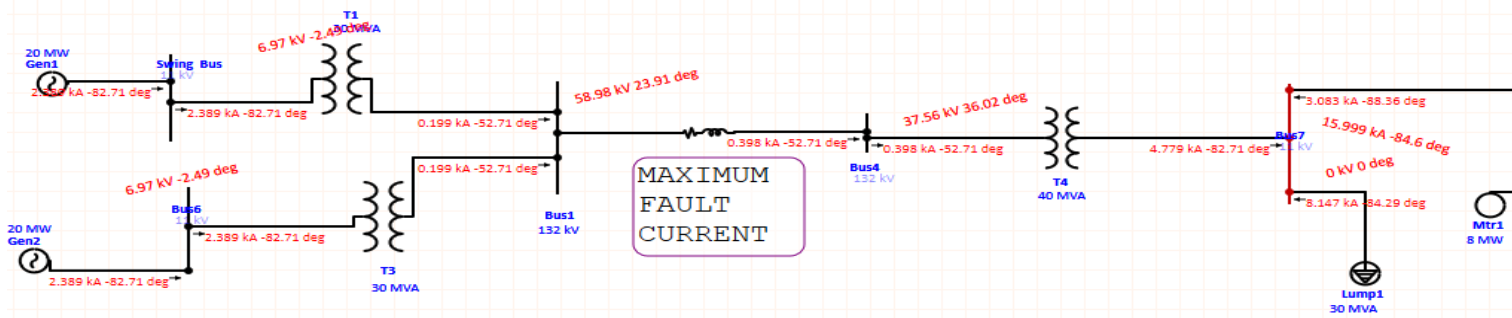
3). BUS 1:-



4). BUS 4:-

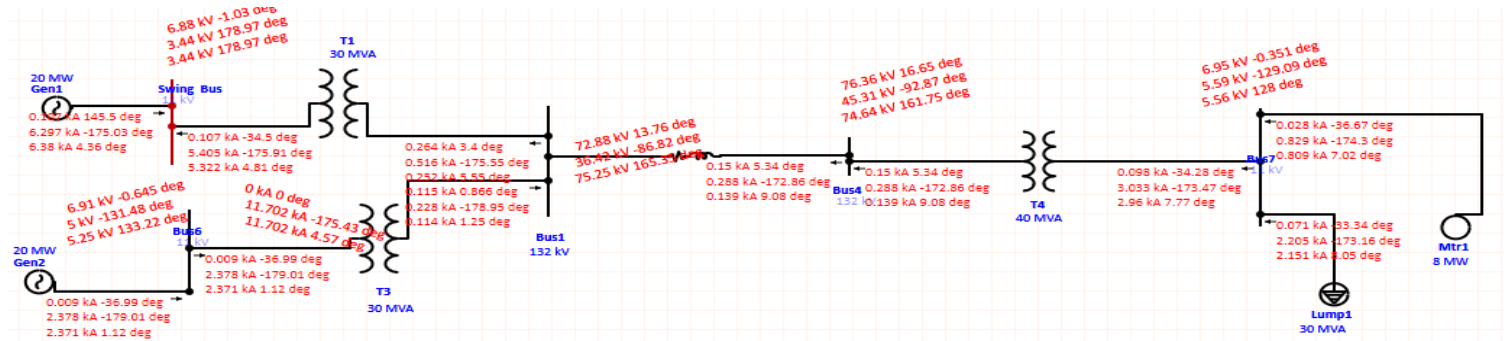


5). BUS 7:-

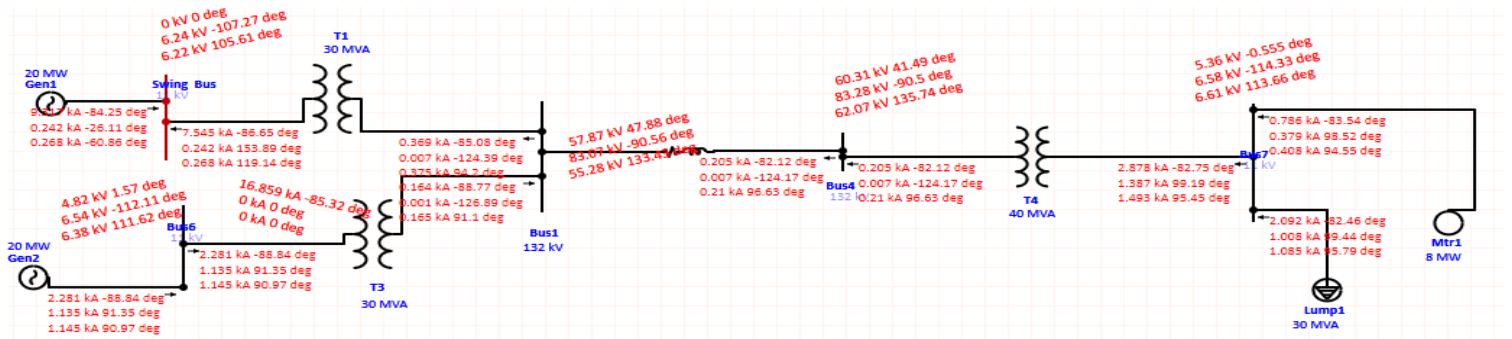


Asymmetrical Faults:-

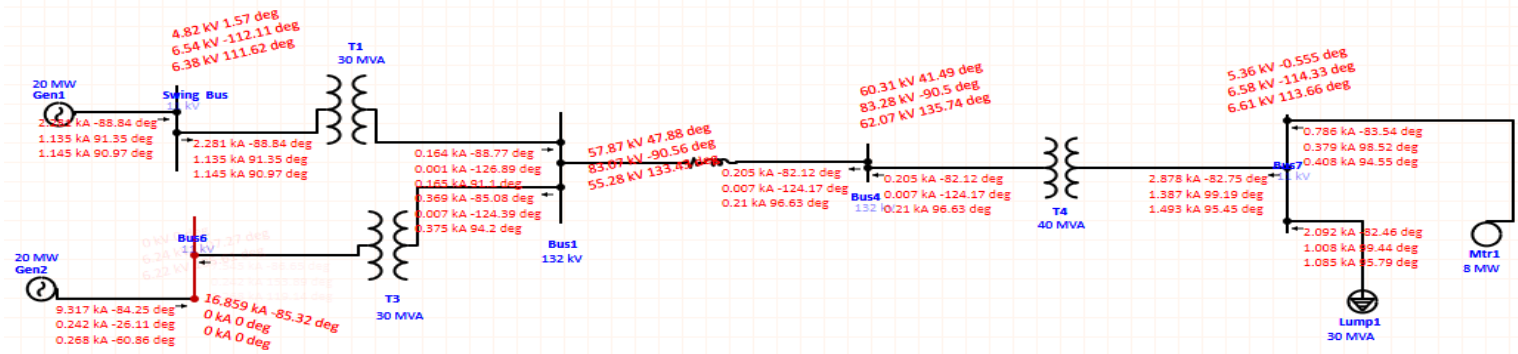
Swing Bus (L-G):-



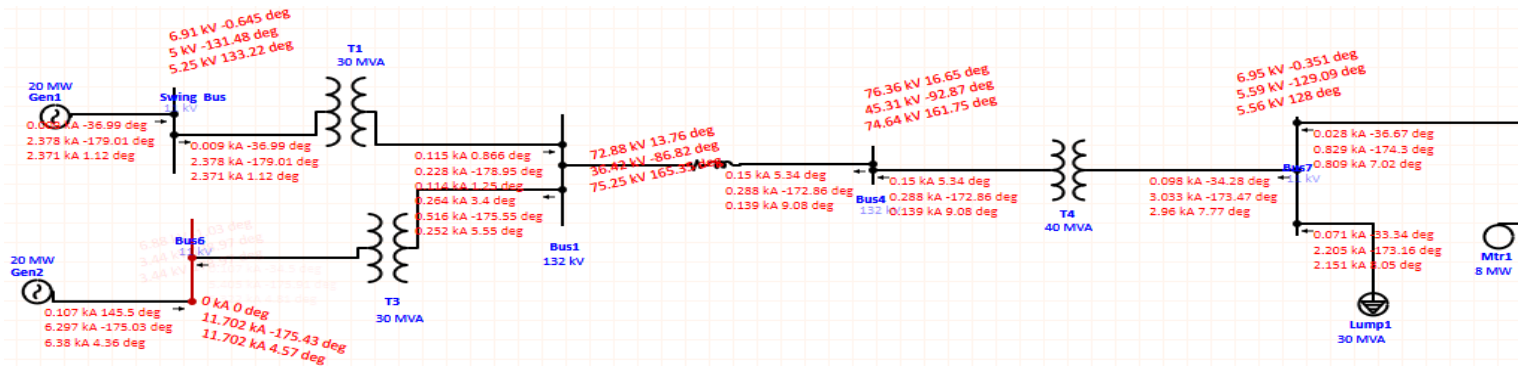
Swing bus (L-L):-



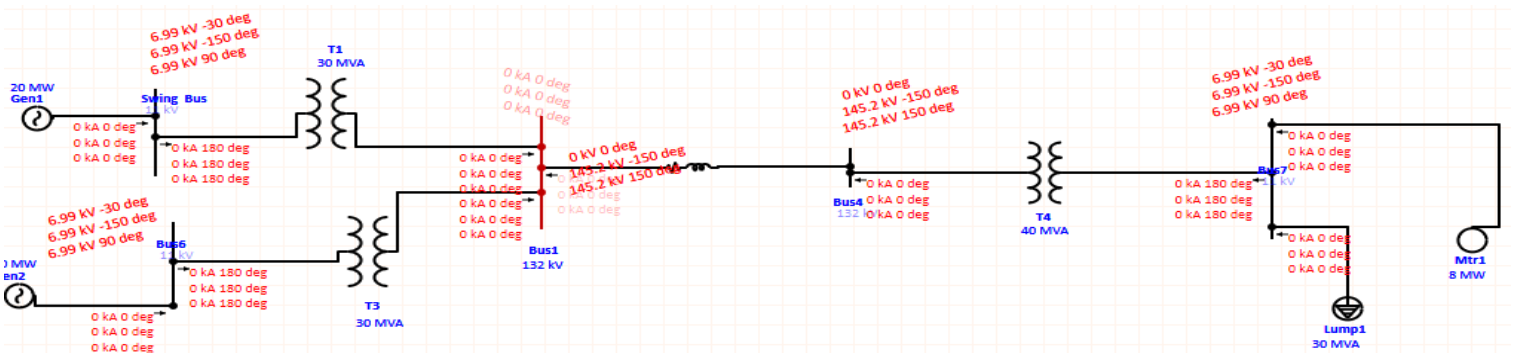
Bus 6 (L-G):-



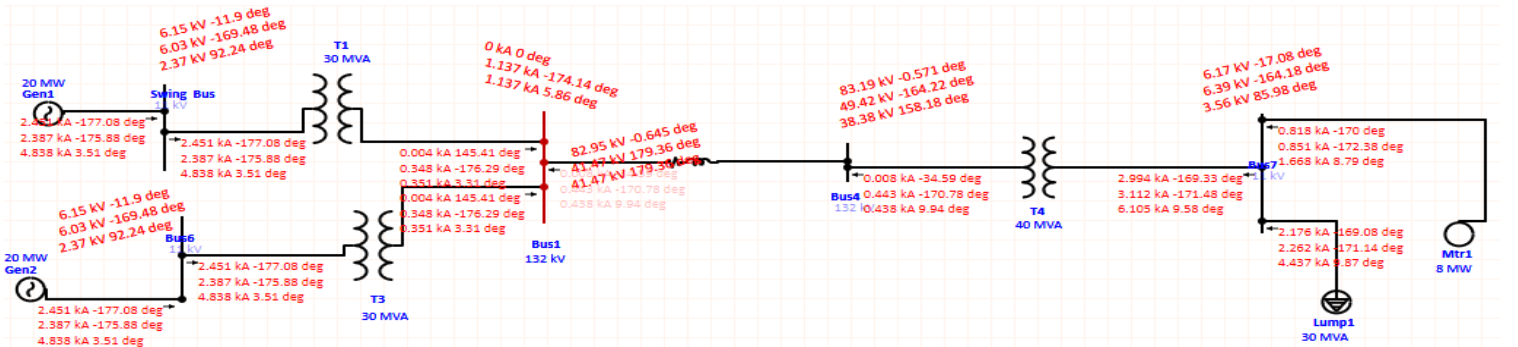
Bus 6 (L-L):-



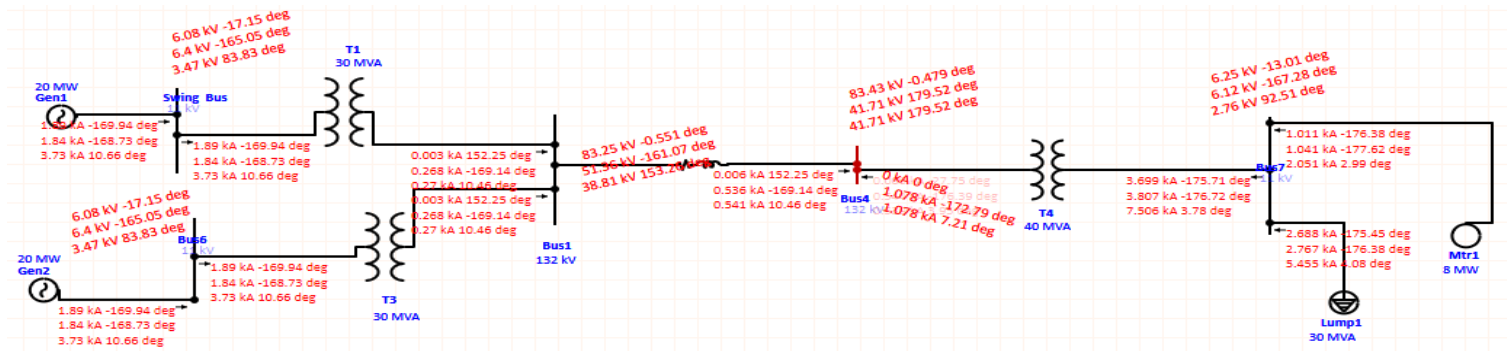
Bus 1 (L-G):-



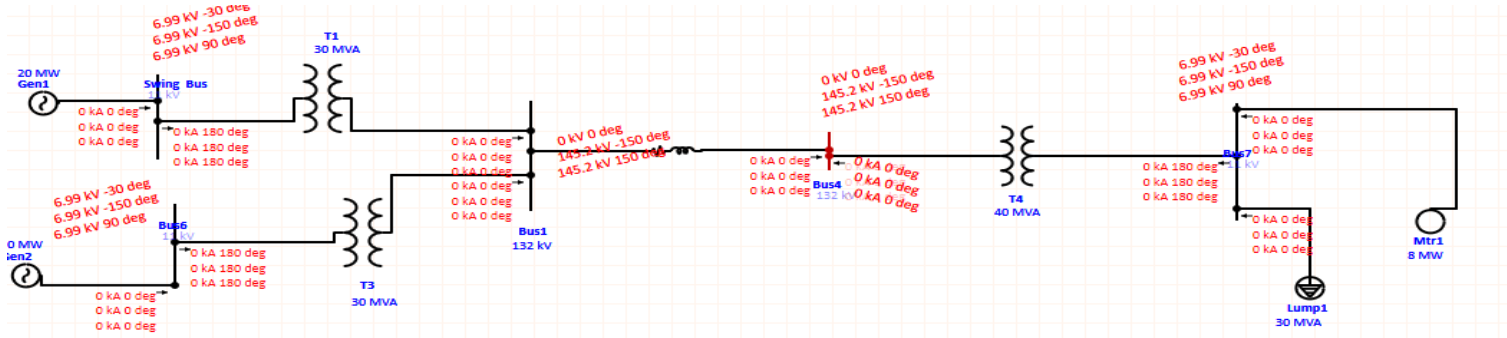
Bus 1(L-L):-



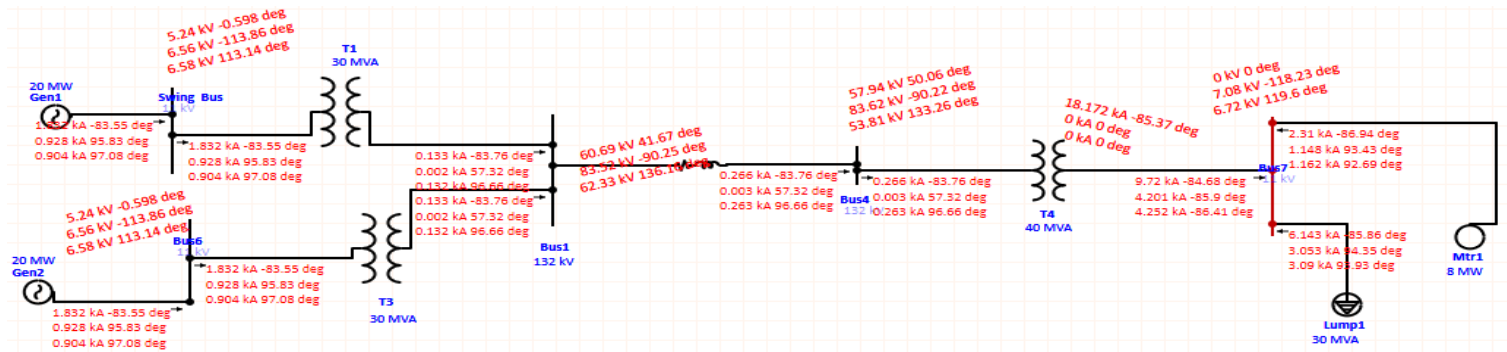
Bus 4 (L-L):-



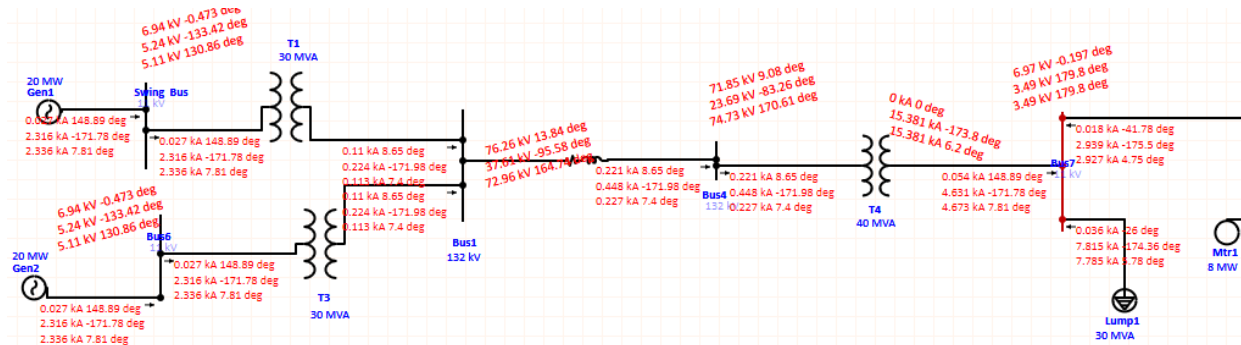
Bus 4(L-G):-



Bus 7(L-G):-



Bus 7(L-L):-



Protection scheme:-

In order to provide protection we take following aspects under consideration

- 1). Reliability.
- 2). Cost
- 3). Selectivity
- 4). Sensitivity
- 5). Speed

Upon inspection of the full load current through load flow analysis and Minimum Fault current via short circuit analysis we select the PS for all relay for OCR as well as Differential protection is made sure by this.

Observation:-

The full load current entering in swing bus is found to be 1094 Amps. The suitable CT ratio chosen for this CT is 1000:5, and the second bus CT ratio for 1200:5 as the load current was 1242 Amps. The differential scheme CT were designed to match the transformation ration.

⇒ $CT_{\text{primary}} = 250:5$ load flow current (203.6 A)

⇒ $CT_{\text{secondary}} = 2500:5$ Load flow current (2444.3 A)

1). Pick up setting for Relay 4 = $1094/1000 \rightarrow 1.094$

2). Pick up setting for Relay 5 = $1242/1200 \rightarrow 1.035$

We use IDMT relay with a curve setting of IEC standard inverse for this application

Overcurrent Relay Editor - Relay4

Info Input Output **OCR** Scheme Logic TCC kA Model Info Checker Remarks Co

ALSTOM
P342

OC Level
OC1 ☒ Enabled ☐ Integrated Curves
☐ Block TOC by IOC & combine for this level

Device Parameters
Selected Device ID Type FLA
Gen1 Generator 1234.97

Phase Ground Sen. Ground

☒ Overcurrent
Curve Type IEC - Standard Inverse Terminal Phase
Pickup Range 0.08 - 4 xCT Sec Multiples
Pickup 1.094 Step: 0.01 Relay Amps 5.47
Time Dial 0.2 Step: 0.025

☒ Instantaneous
Instantaneous Terminal Phase
Pickup Range 0.08 - 10 xCT Sec Multiples
Pickup 5 Step: 0.01 Relay Amps 25
Delay Range 0 - 200 sec
Delay (sec) 0.01 Step: 0.01

☐ Directional 67

Overcurrent Relay Editor - Relay5

Info Input Output **OCR** Scheme Logic TCC kA Model Info Checker Remarks C

ALSTOM
P342

OC Level
OC1 ☒ Enabled ☐ Integrated Curves
☐ Block TOC by IOC & combine for this level

Device Parameters
Selected Device ID Type FLA
Gen2 Generator 1234.97

Phase Ground Sen. Ground

☒ Overcurrent
Curve Type IEC - Standard Inverse Terminal Phase
Pickup Range 0.08 - 4 xCT Sec Multiples
Pickup 1.035 Step: 0.01 Relay Amps 5.175
Time Dial 0.1 Step: 0.025

☒ Instantaneous
Instantaneous Terminal Phase
Pickup Range 0.08 - 10 xCT Sec Multiples
Pickup 5 Step: 0.01 Relay Amps 25
Delay Range 0 - 200 sec
Delay (sec) 0.01 Step: 0.01

☐ Directional 67

Figure 1 The setting of Relay 4 and Relay 5 on GENSET

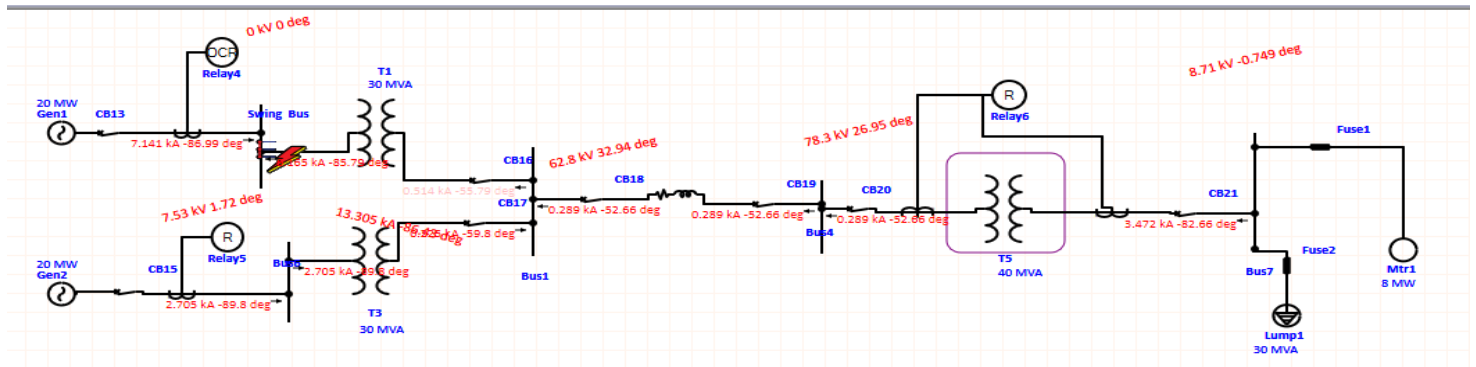
List of protection equipment:-

Component	Manufacturer	Current Rating/Model
CB1& CB2 (Circuit Breaker)	ABB	1250 A/HD4-HPA24
OCR (Relay)	ABB	REG630 (Gen protection)
CB9 & CB 10 (Circuit Breaker)	Hyundai	HEF-630
CB5	ABB	250 A/ 25HKSA1000
CB11	ABB	2500 A/HD4-HPA24
CB3	Hyundai	HEF-630
Relay 6	ABB	RET 541 differential relay
Relay 4 and Relay 5	Alstom	P342 OCR

Following is the schematic of our SLD after applying protection of all sorts (OCR, Differential etc.)

Protection scheme test for selectivity:-

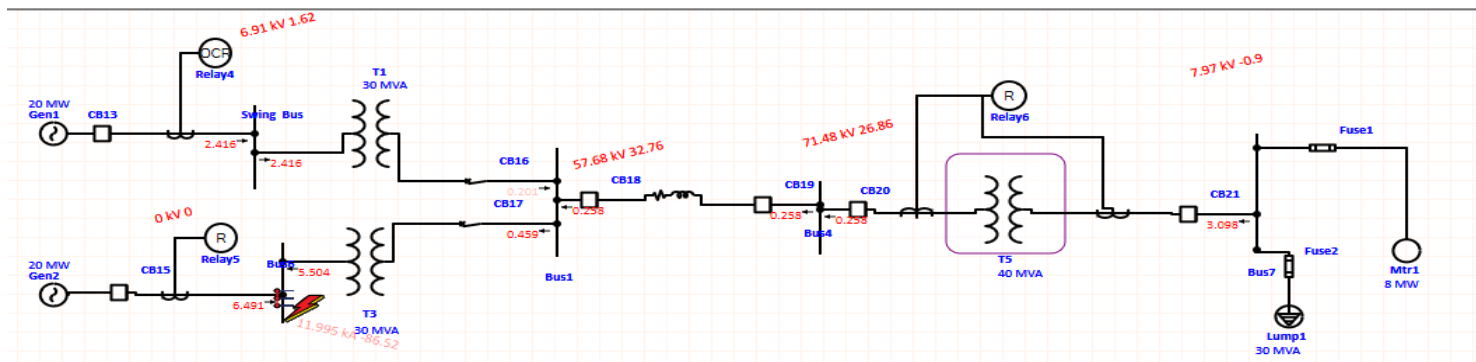
Fault on Swing Bus:-



Sequence-of-Operation Events - Output Report: king

3-Phase (Symmetrical) fault on bus: Swing Bus					
Data Rev.: Base		Config: Normal		Date: 21-01-2023	
Time (ms)	ID	If (kA)	T1 (ms)	T2 (ms)	Condition
10.0	Relay4	7.141	10.0		Phase - OC1 - 50
22.2	Relay4	7.141	< 22.2		Phase - OC1 - 51
57.4	Relay5	2.705	57.4		Phase - OC1 - 51
603	Relay5	2.705	< 603		Overload Phase - Thermal

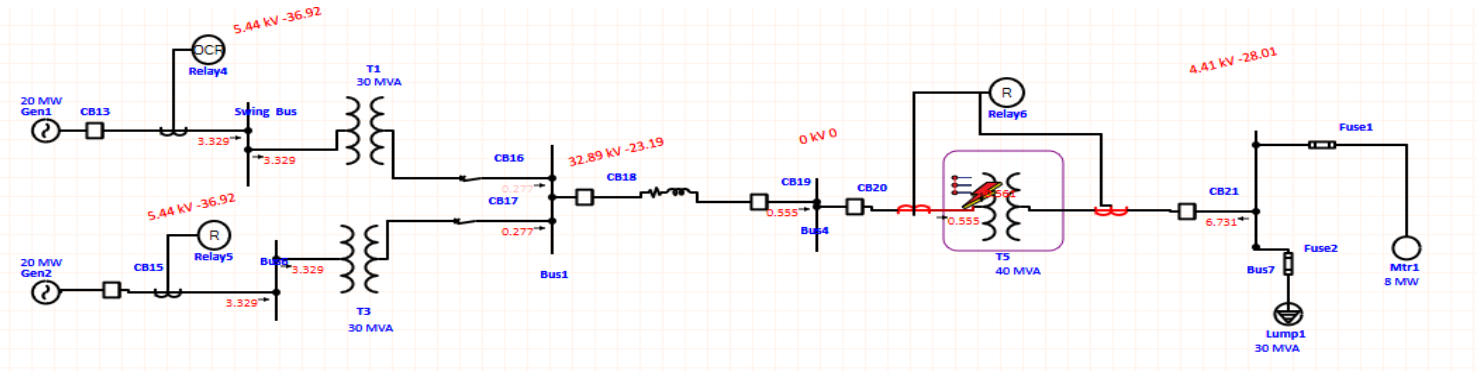
Fault on Bus 6:-



Sequence-of-Operation Events - Output Report: king

3-Phase (Symmetrical) fault on bus: Bus6					
Data Rev.: Base		Config: Normal		Date: 21-01-2023	
Time (ms)	ID	If (kA)	T1 (ms)	T2 (ms)	Condition
20.0	Relay5	6.491	20.0		Phase - OC1 - 50
27.6	Relay5	6.491	27.6		Phase - OC1 - 51
28.8	Relay4	2.416	28.8		Phase - OC1 - 51
603	Relay5	6.491	< 603		Overload Phase - Thermal

Fault on Transformer T2:-



3-Phase (Symmetrical) fault on connector between CT7 & T5. Adjacent bus: Bus4

Data Rev.: Base

Config: Normal

Date: 21-01-2023

Time (ms)	ID	If (kA)	T1 (ms)	T2 (ms)	Condition
20.0	Relay6		20.0		Phase - 87
26.1	Relay4	3.329	26.1		Phase - OC1 - 51
45.5	Relay5	3.329	45.5		Phase - OC1 - 51
603	Relay5	3.329	< 603		Overload Phase - Thermal

Conclusion:-

In short the protection scheme exhibits the selection order what we intend to have such a scheme is useful for both Overcurrent as well as Differential protection.